

RAJAVI GONTIYA

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 [Rajavi Gontiya](#)

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 [Huntsville, AL](#)

EDUCATION

University of Alabama in Huntsville

Master of Science in Software Engineering

Cum GPA: 3.7 Expected: August 2025

SKILLS

Languages & Frameworks:

Python • C++ • Rust (Basic) •
Django/Flask • REST APIs • OAuth

Cloud & DevOps:

Azure Data Lake • Azure Data Factory •
Azure SQL • Azure Monitor • Docker •
CI/CD Pipelines • Azure Cloud Platform
Databricks

Security & Standards:

Secure Coding Practices • OWASP • Web
Application Security • API
Authentication/Authorization • CodeQL •
Cybersecurity Compliance

Tools & Technologies:

Pandas • NumPy • PySpark • MongoDB •
MySQL • PostgreSQL • HTML • Scikit-
learn • JSON • Git • JIRA • Unix/Linux •
Windows • Elastic Search • OpenStack

Methodologies:

Agile • Test-Driven Development •
Modular Software Development • Version
control systems

PROFESSIONAL CERTIFICATION

- **Microsoft Certified: [Azure Data Fundamentals](#)**
- Secure Coding in Python ([Coursera](#))

COURSEWORK

Data Structure, Advanced Operating System, Object Oriented Technology (C++, JAVA), Analysis and Design of Algorithm, Cloud Computing, Machine Learning, Software Engineering Processes, Software Reverse Engineering

AREA OF INTEREST

Cloud Computing / Secure Software Development/ Software Engineering / Web & API Security/ Data Processing/ Application Security/ Application development/ Agile methodologies

WORK EXPERIENCE

Graduate Research Assistant | University of Alabama in Huntsville

Aug 2023 – Present

- Thesis-driven research project evaluating the code generation capabilities of leading **Generative AI models**: ChatGPT (GPT-4 mini), GitHub Copilot, Code LLaMA 3.3, and DeepSeek Web.
- Developed an extensible framework that uses open-source tools such as **statsmodels, pingouin, scipy, and matplotlib** to statistically evaluate AI-generated Python code.
- By applying rigorous statistical methods, including repeated measures **ANOVA**, post-hoc Tukey HSD, and sphericity tests, this thesis quantifies the strengths and weaknesses of each AI model.

Azure Data Engineer | Infosys Pvt. Ltd., India

Aug 2021 – July 2023

- Built and orchestrated data pipelines using **Azure Data Factory** to extract, transform, and load (**ETL**) large datasets from multiple sources into **Azure Data Lake and Azure SQL**.
- Designed data ingestion workflows to move structured and unstructured data into **Azure Data Lake**, ensuring compliance with data governance and security standards.
- Automated data transformation jobs using **Databricks** notebooks (**PySpark**) for high-volume data processing, improving pipeline runtime efficiency.
- Integrated **MongoDB** and other **NoSQL** sources into Azure workflows for real-time and batch processing scenarios.
- Scheduled and monitored data workflows using **ADF triggers** and alerts, reducing failure rates through proactive logging and error handling.

Software Developer | Cyber Infrastructure Pvt. Ltd., India

Feb 2018 – July 2021

- Designed and developed scalable **RESTful APIs** using **Python** and **Flask/Django**, enabling seamless communication between microservices in production-grade applications.
- Built event-driven microservices using **Celery** for background task processing.
- Integrated **ElasticSearch** for real-time search functionality and analytics, enabling users to query large datasets with near-instant feedback.
- Contributed to **CI/CD** pipelines and version control workflows using **Git**, GitHub/GitLab, and **Docker**, ensuring smooth code deployment and rollback processes.

PROJECT EXPERIENCE

Analyzing Code Generation by AI Models: An Anova-Based Study of Quality, Consistency, and Composite Ranking

- Published in ProQuest [Link](#)
- Determine **statistically significant** differences across Generative AI models and coding tasks. This contributes a reusable research tool for future benchmarking studies.

Multiple Disease Prediction based on Health Indicators | Python, Seaborn, Pandas, Scikit-learn, PyTorch, TensorFlow

- Developed a machine learning-based prediction system using **Python, Seaborn, and Pandas**, achieving an **89% accuracy rate** across multiple disease categories.
- Applied **Scikit-learn, PyTorch, and TensorFlow** to build, train, and compare multiple classification models.
- Implemented advanced **data preprocessing, clustering, and hyperparameter tuning** to enhance model performance.
- Trained and evaluated multiple **Machine Learning models** (e.g., Decision Trees, Random Forests, SVM, etc.) on each cluster to identify the best-performing algorithm.